4. Estimation and Test of Hypothesis

Statistical Inference : divided into 2 parts

1) Estimation

2) Test of Hypothesis

Max error of Estimates.

$\frac{x-\mu}{\frac{\sigma}{\sqrt{n}}}$ will become a random variable;

$$-z_{\alpha/2} \leq \frac{x-\mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{\alpha/2}$$

$$\frac{|x-\mu|}{\frac{\sigma}{\sqrt{n}}} \leq z_{\alpha/2}$$

MEANS

1) Maximum error of estimates.

$$E = \begin{cases} z_{\alpha/2} \frac{\sigma}{\sqrt{n}} & \text{when } n \geq 30 \text{ (large sample)} \\ t_{\alpha/2} \frac{s}{\sqrt{n}} & \text{when } n < 30 \text{ (small sample)} \end{cases}$$

2) Sample Size (n)

$$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \Rightarrow n = \left(z_{\alpha/2} \cdot \frac{\sigma}{E}\right)^2$$

3) Confidence Interval

$$\bar{x} - E < \mu < \bar{x} + E$$

4) Degree of confidence. → Same formula we can use.

NOTE: Mostly used $(1-\alpha)100\%$ values are 99% & 95%

$$\left.\begin{aligned} z_{0.025} &= 1.96 \\ z_{0.005} &= 2.576 \text{ or } 2.58. \end{aligned}\right\}$$

t-distribution.
not z-distribution
as there is $\alpha/2$

Q. A random sample of size 100 has a SD of 5. What can you say about max error with 95% confidence

sample size $\Rightarrow n = 100$ (large)

sample SD $\Rightarrow s = 5$

95% confidence

$\alpha = 5\% \Rightarrow \alpha/2 = 2.5\% = 0.025$

$Z_{\alpha/2} = Z_{0.025} = 1.96$

Now, max error of estimate $\Rightarrow E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$= Z_{\alpha/2} \frac{s}{\sqrt{n}}$ $\because \sigma$ is unknown

$= 1.96 \times \frac{5}{\sqrt{100}}$

$E = 0.98$

Q: What is the max error one can except to make with probability 0.9 when using the mean of sample of size 64. To estimate the mean of population with $\sigma^2 = 256$.

Sample size $\Rightarrow n = 64$ (large)

confidence $= 90\%$ (0.9)

Error $\alpha = 10\% = 0.1$

$\alpha/2 = 0.05$

$Z_{\alpha/2} = Z_{0.05} = 1.645$.

$\sigma^2 = 256 \Rightarrow \sigma = 16$.

$E = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 1.645 \times \frac{16}{\sqrt{64}}$

$= 3.29$

Q: Assuming that $\sigma = 20$, how large the random sample taken to assert with probability 0.95 that the sample mean will not differ from the true mean by more than 3.0 points.

$\sigma = 20$

confidence = 0.95

$\alpha = 0.05$

$\alpha/2 = 0.025$

$z_{\alpha/2} = z_{0.025} = 1.96$

$E = 3.0$

$n = ?$

$$n = \left(z_{\alpha/2} \frac{\sigma}{E}\right)^2 = \left(1.96 \times \frac{20}{3}\right)^2$$

$= (170.7377778).$

$\simeq 171$

Q: It is desired to estimate the mean no. of hours of continuous use until a certain computer will 1st require repairs. If it can be assumed that $\sigma = 48$ hours, how large a sample be needed so that I will be able to assert with 90% confidence that the sample mean is off by atmost 10 hours.

$\sigma = 48$

confidence = 90%

$\alpha = 10\% \Rightarrow \alpha/2 = 0.05$

$z_{\alpha/2} = z_{0.05} = 1.645$

$E = 10$

$$n = \left(z_{\alpha/2} \frac{\sigma}{E}\right)^2 = \left(1.645 \times \frac{48}{10}\right)^2$$

$= 62.346816$

$\simeq 62.$

Q: A random sample of size 81 was taken whose variance is 20.25 and mean is 32. Construct 98% confidence interval

$n = 81 \Rightarrow$ large

$s^2 = 20.25 \Rightarrow s = 4.5$

mean $\bar{x} = 32$

confidence = 98%.

$\alpha = 2\% = 0.02$

$\alpha/_2 = 0.01$

$z_{\alpha/_2} = z_{0.01} = 2.326.$

$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$= z_{\alpha/2} \frac{s}{\sqrt{n}} = 2.326 \times \frac{4.5}{\sqrt{81}}$

$= 1.163.$

confidence Interval $= \bar{x} - E < \mu < \bar{x} + E$

$= 30.837 < \mu < 33.163$

Q: A random sample of size 100 is taken from a population with $\sigma = 5.1$ and the sample mean $\bar{x}$ is 21.6. Construct 95% confidence interval for the pop mean μ

$n = 100 \Rightarrow$ large

$\sigma = 5.1$

confidence = 95%

$\alpha = 5\%$

$\alpha/_2 = 2.5\% = 0.025$

$z_{\alpha/_2} = z_{0.025} = 1.96$

$\bar{x} = 21.6$

$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{5.1}{\sqrt{100}}$

$= 0.9996$

confidence interval

$= \bar{x} - E < \mu < \bar{x} + E$

$= 20.6004 < \mu < 22.5996$

Q: Find 95% confidence limits for the mean of a normally distributed population from which the sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14

$x = 15, 17, 10, 18, 16, 9, 7, 11, 13, 14$

$n = 10 \Rightarrow$ small.

sample mean $\bar{x} = \frac{\Sigma x_i}{n}$

$= 13.$

$$SD = s = \sqrt{\frac{\Sigma(x_i - x)^2}{n-1}} = \sqrt{\frac{4+16+9+25+9+16+36+4+0+1}{9}}$$

$= 3.65$

95% confidence $\Rightarrow \alpha = 5\%.$

$\alpha/2 = 0.025$

$t_{0.025}, \nu = 9$

$t_{0.025, 9} = 2.262$

$$E = t_{\alpha/2} \frac{s}{\sqrt{n}} = 2.262 \times \frac{3.65}{\sqrt{10}}$$

$= 2.6108 = 2.611$

confidence Interval $= \bar{x} - E < \mu < \bar{x} + E$

$= 13 - 2.611 < \mu < 13 + 2.611$

$= 10.389 < \mu < 15.611.$

Q: The mean of the random sample is an unbaised estimate of the mean of the population 3, 6, 9, 15, 27

(i) List all the possible samples of size 3 that can be taken without replacement from the finite population

(ii) Calculate the mean of the each sample lised in (i) and assigning each sample probability of $1/10$

Verify that the mean of these $\bar{x}$ is equal to 12 which is equal to the mean of the population θ i.e., $E(\bar{x}) = \theta$ i.e, prove that $\bar{x}$ is unbaised estimator of θ.

x_i's $\rightarrow 3, 6, 9, 15, 27$, $N = 5$

$n = 3$

pop mean $\mu = \frac{\Sigma x_i}{N} = \frac{3+6+9+15+27}{5} = 12$

No. of samples of size 3 that can be formed without replacement is $k = {}^5C_3 = 10$

(3,6,9) (3,6,15) (3,6,27)
(3,9,15) (3,9,27) (3,15,27)
(6,9,15) (6,9,27) (6,15,27)
(9,15,27)

Sampling distribution

6, 8, 12
9, 13, 15
10, 14, 16
17

$\bar{x}$'s.

$\bar{x}$	6	8	9	10	12	13	14	15	16	17
$f(x)$	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10	1/10

True mean $= \Sigma \bar{x} f(x)$

$E(x) = 6 \times \frac{1}{10} + 8 \times \frac{1}{10} + \cdots + 16 \times \frac{1}{10} + 17 \times \frac{1}{10}$

$= \frac{1}{10}[6+8+9+10+12+13+14+15+16+17]$

$= 12$

$\therefore E(\bar{x}) = 12 =$ pop mean

Hence $\bar{x}$ is unbiased estimator of θ

Q A random sample of 10 boys had the following IQ's 70, 120, 110, 101, 88, 83, 95, 98, 107, & 100. find a reasonable range in which most of the mean IQ values of samples of 10 boys lie.

$n = 10 \Rightarrow$ small.

$x = 70, 120, 110, 101, 88, 83, 95, 98, 107, 100$

$$\text{mean} = \frac{\sum x_i}{n} = 97.2$$

$$SD = s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{739.84 + 519.84 + 163.84 + 14.44 + 84.64 + 201.64 + 4.84 + 0.64 + 96.04 + 7.84}{9}}$$

$$= 14.273$$

If confidence is not mentioned use 95% or 99%.

let us take here 99% confidence

$\alpha = 1\% \Rightarrow \alpha/2 = 0.5\%$

$\cancel{z_{\alpha/2} = z_{0.005}}$

$t_{0.005}$ at $9 = 3.25$

$$E = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$= 3.25 \times \frac{14.273}{\sqrt{10}} = \cancel{14.443}\ 14.668$$

Confidence Interval is

$$82.53 < \mu < 111.868.$$

Degree of confidence problems (means).

Q The efficiency expert of a comp company tested 40 engineers to estimate the avg time it takes to assemble a certain comp component getting a mean of 12.73 min & SD of 2.06 min

i) If $\bar{x} = 12.73$ is used as point estimate of the actual avg time to perform the task. determine the max error with 99% confidence

(ii) Construct 98% confidence interval for the true avg time it takes to do the job

(iii) with what confidence can we assert that the sample mean doesn't differ from true mean by more than 30 sec.

sample size $n = 40$

sample mean $\bar{x} = 12.73$ min

sample standard deviation $s = 2.06$ min

(i) 99% confidence

$$z_{\alpha/2} = z_{0.005} = 2.576$$

$$E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$= 2.576 \times \frac{2.06}{\sqrt{40}}$$

$$= 0.839$$

(ii) 98% confidence.

$\alpha/2 = 0.01$ $z_{\alpha/2} = 2.326$

$$E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 2.326 \cdot \frac{2.06}{\sqrt{40}}$$

$$= 0.758$$

Interval $\Rightarrow 12.73 - 0.758 < \mu < 12.73 + 0.758$

$$\Rightarrow 11.972 < \mu < 13.488$$

(iii) $E = 30$ sec

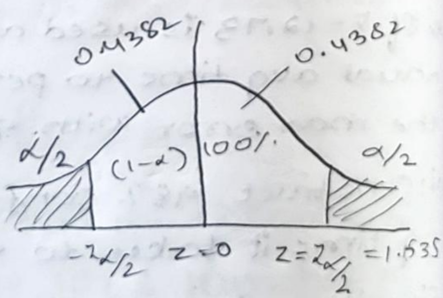
$= 30/60$ min $= 0.5$ min

$$E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$0.5 = z_{\alpha/2} \cdot \frac{2.06}{\sqrt{40}}$$

$$z_{\alpha/2} = \frac{0.5 \times \sqrt{40}}{2.06}$$

$$z_{\alpha/2} = 1.535$$



From fig, $P(0 < z < 1.535)$

$$\int_0^{1.54} f(z)dz = 0.4382$$

$\therefore$ Degree of confidence $= 0.4382 + 0.4382$

$= 0.8764 \simeq 88\%$

Q: In a study of automobile insurence a random sample of 80 body repair costs had a mean of ₹472.36 and SD of ₹62.35. If $\bar{x}$ is used as a point estimate to the true avg repair cost with what confidence we can assert that the max error doesn't exceed ₹10.

$E = 10.$

$\bar{x} = 472.36$

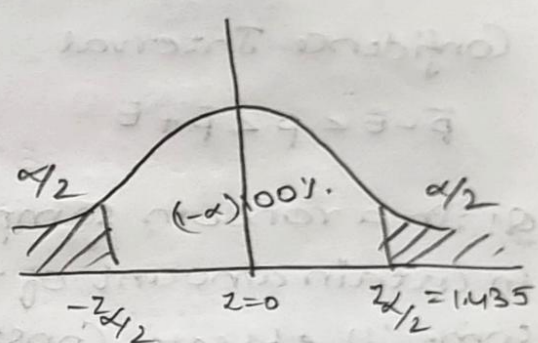
$s = 62.35.$

$n = 80$ (large).

$$E = z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$z_{\alpha/2} = \frac{10 \times \sqrt{80}}{62.35}$$

$= 1.4345$



from fig $P(0 < z < 1.435)$

$$\int_0^{1.44} f(z)dz = 0.4251$$

$\therefore$ Degree of confidence $= 0.4251 + 0.4251$

$= 0.8502$

$\simeq 85\%$.

Proportions

$\bar{p}$ - sample proportion

p - population proportion

$q = 1 - p$

$\bar{p} = x/n$

$\bar{q} = 1 - \bar{p}$

Maximum Error.

$$E = \begin{cases} z_{\alpha/2}\sqrt{\dfrac{\bar{p}\bar{q}}{n}} & \text{for large sample} \\ t_{\alpha/2}\sqrt{\dfrac{\bar{p}\bar{q}}{n}} & \text{for small sample} \end{cases}$$

Sample size.

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \bar{p}\bar{q}$$

Confidence Interval.

$$\bar{p} - E < p < \bar{p} + E$$

Q: In a random sample of 160 workers exposed to certain amount of radiation 24 experienced some ill effects. Construct 99% confidence interval for the corressponding true percentage (population proportion).

Note: 1) If $\bar{p}$ and $\bar{q}$ are not given then take $\bar{p} = \bar{q} = 1/2$

2) If degree of confidence is not given then in proportions replace $z_{\alpha/2}$ with 3.

x = No. of people experiencing ill effects of radiation

$x = 24$, $n = 160$ (large)

Sample proportion $\bar{p} = \dfrac{x}{n} = \dfrac{24}{160} = \dfrac{3}{20} = 0.15$

$\bar{q} = 0.85$

99% confidence

$z_{\alpha/2} = z_{0.005}$

$= 2.576$

max error $E = z_{\alpha/2}\sqrt{\frac{\bar{p}\bar{q}}{n}}$

$= 2.576\sqrt{\frac{0.15 \times 0.85}{160}}$

$= 0.0727$

Confidence interval $\Rightarrow \bar{p} - E < p < \bar{p} + E$

$= 0.0773 < p < 0.2227.$

Q: What is the size of the smallest sample req to estimate an unknown proportion to within a maximum error of 0.06 with atleast 95% confidence.

$E = 0.06$

$\bar{p} = \bar{q} = 0.5$

confidence $= 95\%$.

$\alpha = 5\% \Rightarrow \alpha/2 = 2.5\%$.

$z_{\alpha/2} = z_{0.025} = 1.96$

$n = \left(\frac{1.96}{0.06}\right)^2 0.5 \times 0.5$

$= 266.777$

$\simeq 267.$

Q: In a sample of 100 packages shipped by air fright, 13 had some damage. Construct 95% confidence interval for the true proportion of damage package.

X = damaged package.

$x = 13$, $n = 100$ (large)

proportion $= \bar{p} = \frac{13}{100} = 0.13$

$\bar{q} = 1 - \bar{p} = 0.87$

confidence $= 95\%$.

$\alpha = 5\% \Rightarrow \alpha/2 = 2.5\% = 0.025$

$z_{\alpha/2} = z_{0.025} = 1.96.$

$E = z_{\alpha/2}\sqrt{\frac{\bar{p}\bar{q}}{n}} = 1.96\sqrt{\frac{0.13 \times 0.87}{100}}$

$= 0.0659$

confidence interval = $\bar{p} - E < p < \bar{p} + E$

$0.0641 < p < 0.1959$.

(continuation of date of confidence problems)

Q A sample of 11 rats from a central population has an avg blood viscosity of 3.92 with a SD of 0.61 estimate 95% confidence limits for the mean blood viscosity of the population

$n = 11$ (small)

$\bar{x} = 3.92$

$s = 0.61$

confidence = 95%.

$\alpha = 5\%$. $\alpha/2 = 0.025$

since sample is small we find $t_{\alpha/2, \nu}$

$\nu = 10$

$t_{0.025, 10} = 2.228$

$E = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 2.228 \times \frac{0.61}{\sqrt{11}}$

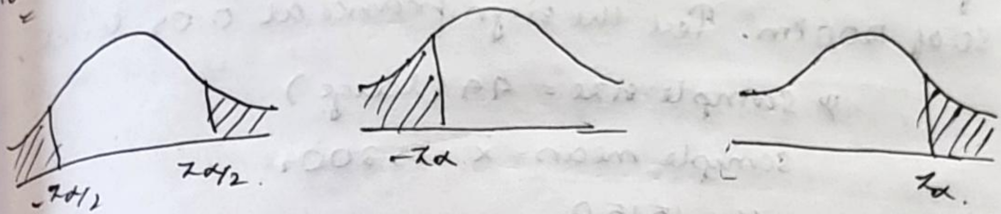
$= 0.409$

confidence interval $\Rightarrow \bar{x} - E < \mu < \bar{x} + E$

$= 3.511 < \mu < 4.329$

Test of Hypothesis

Two tailed　　　left tailed　　　right tailed



- The region where we accept null hypothesis then we call acceptance region
- The region where we reject null hypothesis then we call rejection region.

Q. An ambulance service claims that it takes on an avg less than 10 min to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 min and the var of 16 min. Test the significance at 0.05 level.

1) sample size $n = 36$ (large)

sample mean $\bar{x} = 11$

$var = s^2 = 16 \Rightarrow s = 4$

$z_{0.05} = 1.645$

$z_{0.005} = 2.576$

$z_{0.025} = 1.96$

$z_{0.01} = 2.326$

2) N.H $H_0 : \mu = 10$.

A.H $H_0 : \mu < 10$

3) $LOS = \alpha = 0.05$

$z_\alpha = z_{0.05} = 1.645$

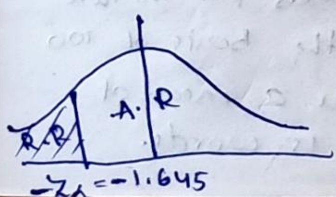
4) Test of Hypothesis : $z = \dfrac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ under N.H.

$$z = \frac{11-10}{\frac{4}{\sqrt{36}}} = \frac{1 \times 6}{4}$$

$= 3/2$

$= 1.5$

5) conclusion



Since the value of z falls under A.R, accept N.H

$\Rightarrow$ His claim is false.

Q. It is claimed that a random sample of 49 tyres has a mean lyf of 15,200 km. This sample was drawn from a population whose mean is 15150 km and a SD of 1200 km. Test the significance at 0.05 level.

1) sample size = 49 (large)

sample mean = $\bar{x} = 15200$.

$\mu = 15150$

$\sigma = 1200$

2) N.H H_0: The given sample is taken from pop having mean $\mu = 15150$.

A.H H_1: It is not taken from this population, $\mu \neq 15150$

TTT → two tailed test.

3) Level of significance (LOS) ⇒ $\alpha = 0.05$

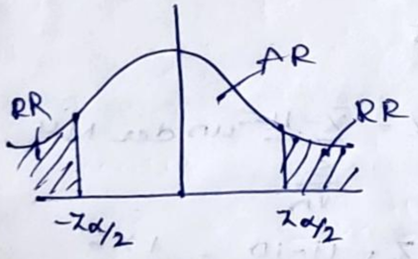
$\alpha/2 = 0.025$

4) Test statistic ⇒ $z = \dfrac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ under null hypothesis

$$z = \dfrac{15200 - 15150}{\frac{1200}{\sqrt{49}}}$$

$= 0.29166$

5) conclusion



Accept null Hypothesis so his claim is true.

Q. A lady stenographer claims that she can take dictation at the rate of 120 words per minute can we reject her claim on the basis of 100 trails in which she demonstrates a mean of 116 words and a std. dev of 15 words.

1) sample size $n = 100$ (large)
sample mean $\bar{x} = 116$
sample SD $s = 15$
$\mu = 120$

2) N.H $H_0: \mu = 120$
A.H $H_1: \mu < 120$ or $\mu \neq 120$
LTT TTT

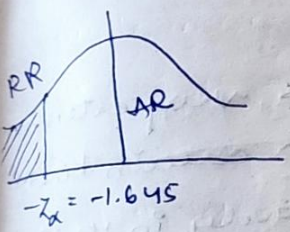
3) LOS let $\alpha = 0.05$
$z_\alpha = z_{0.05} = 1.645$

4) Test statistic $z = \dfrac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ under NH.

$$z = \frac{116 - 120}{\frac{15}{\sqrt{100}}}$$

$$= \frac{4 \times 10}{15} = -2.6$$

5) Conclusion



Since the value of z falls under RR, Reject N.H
$\Rightarrow$ Her claim is false.

Test of Hypothesis of single mean and small sample.

Q: The heights of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it resonable to believe that the avg height greater than 64 inches? Test at 5% level of significance.

1) $n = 10$ (small)

Sample mean $= \dfrac{\Sigma x_i}{n} = \dfrac{660}{10} = 66$

$$SD = s = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{90}{9}} = \sqrt{10} = 3.162.$$

2) NH $H_0: \mu = 64$

AH $H_1: \mu > 64$ (RTT).

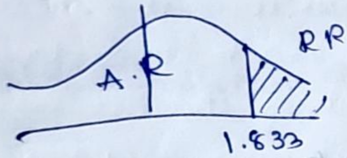
3) LOS $\alpha = 0.05$

$t_{\alpha,9} = t_{0.05,9} = 1.833$.

4) Test statistics $t = \dfrac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ under NH

$$t = \frac{66 - 64}{\frac{\sqrt{10}}{\sqrt{10}}} = 2.$$

5) Conclusion



Since the value of t falls under RR, so rejecting NH

So claim is true $\Rightarrow \mu > 64$.

A random sample from company's very extensive files shows that the orders for a certain kind of machinery were filled respectively in 10, 12, 19, 14, 15, 18, 11 & 13 days. Use the level of significance $\alpha = 0.01$ to test the claim on the avg such orders are filled in 10.5 days.

1) $n = 8$ (small)

Sample mean $= \bar{x} = \dfrac{\Sigma x_i}{n} = \dfrac{112}{8} = 14$.

Sample SD $= s = \sqrt{\dfrac{\Sigma(x_i - \bar{x})^2}{n-1}} = \dfrac{72}{7} = 3.207$

2) N.H $H_0: \mu = 10.5$ (claim).

A.H $H_1: \mu > 10.5$ (RTT)

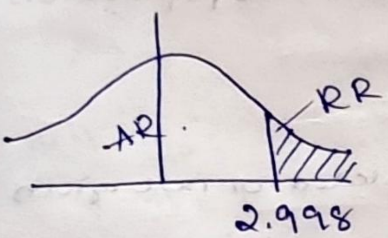
3) LOS $\alpha = 0.01$

$t_{\alpha,\nu} = t_{0.01,7} = 2.998$

4) Test statistics $t = \dfrac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$ under NH

$$t = \frac{14 - 10.5}{\frac{3.207}{\sqrt{8}}} = 3.086.$$

5) Conclusion.



Since the value of t falls under RR, rejecting null hypothesis

⇒ this claim is false.

Q. A random sample of 6 steel beams has a mean compressive test of 58392 PSI and SD of 648 PSI. Use this information and level of significance $\alpha = 0.05$ to test whether the true avg compressive strength of the steel from which this sample come is 58000 PSI. Assume normality

1) $n = 6$ (small)

$\bar{x} = 58392$

$s = 648$

$\mu = 58000$

2) NH H_0: $\mu = 58000$ (belongs)

AH H_1: $\mu \neq 58000$ (doesn't belong)

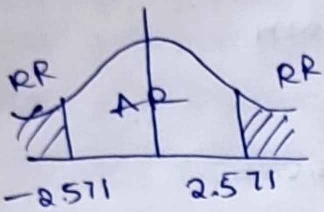
(TTT)

3) LOS $\alpha = 0.05 \Rightarrow \alpha/2 = 0.025$

$t_{\alpha/2,\nu} = t_{0.025,5} = 2.571$

4) Test statistics $\Rightarrow t = \dfrac{\bar{x}-\mu}{\frac{s}{\sqrt{n-1}}} = \dfrac{58392-58000}{\frac{648}{\sqrt{6-1}}} =$ [illegible]

5) Conclusion



Since the value of t falls under AR, we accept NH

$\Rightarrow$ so, the sample is taken from this population only with $\mu = 58000$ PSI.

**
$$t = \begin{cases} \dfrac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} & \text{where s is calculated or not given} \\ \dfrac{\bar{x}-\mu}{\frac{s}{\sqrt{n-1}}} & \text{when s is given.} \end{cases}$$

Test of Hypothesis for difference b/w 2 means large samples.

Test statistic $\Rightarrow$ $$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$ under N.H

N.H $\mu_1 = \mu_2$

A.H $\mu_1 < \mu_2$ L.T.T

$\mu_1 > \mu_2$ RTT

$\mu_1 \neq \mu_2$ T.T.T

Q In a survey of buying ~~horabbits~~ habits, 400 ~~ma~~ women shopper are choosen at random at supermarket A located at ~~er~~ certain section of city. There avg weekly food expenditure is ₹250 with SD of ₹40. For 400 women shoppers choosen at random in supermarket B in another section of city. Avg weekly food expenditure is ₹220 with a SD Rs. 55. Test at 1% level of significance whether the avg weekly food expenditure of 2 population of shoppers are equal.

1) | A | B |
|---|---|
| $n_1 = 400$ | $n_2 = 400$ |
| $\bar{x}_1 = 250$ | $\bar{x}_2 = 220$ |
| $s_1 = 40$ | $s_2 = 55$ |
| μ_1 | μ_2 |

2) NH $H_0 : \mu_1 = \mu_2$ (equal or no diff) ✓

AH $H_1 = \mu_1 \neq \mu_2$ (not equal or some diff) T.T.T.

3) L.O.S $\alpha = 0.01$

$\alpha/2 = 0.005$

$z_{\alpha/2} = z_{0.005} = 2.576$

4) Test statistic

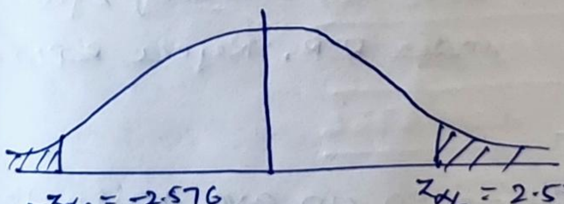
$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$ under NH.

($\because \sigma_1$ & σ_2 are not given replace them with s_1 & s_2)

$$= \frac{(250-220) - (0)}{\sqrt{4 + 7.5625}} = \frac{30}{\sqrt{11.5625}}$$

$$= 8.82$$

5) conclusion



there is significance diff b/w the food expenditure of 2 population

Q: A simple sample of height of 6400 Englishmen has a mean height of 67.85 inch and a SD of 2.56 in while a simple sample of heights of 1600 Australians has a mean of 68.55 inches and a SD of 2.52 inches. Do the data indicate that Australians on an avg taller than Englishmen?

1) Englishmen | Australians

$n_1 = 6400$ (large) | $n_2 = 1600$ (large)

$\bar{x}_1 = 67.85$ | $\bar{x}_2 = 68.55$

$s_1 = 2.58$ | $s_2 = 2.52$

μ_1 | μ_2

2) N.H. $H_0 : \mu_1 = \mu_2$

A.H $H_1 : \mu_1 < \mu_2$ (L.T.T) (claim)

3) L.O.S let $\alpha = 1\% = 0.01$

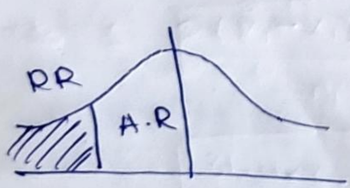
$$Z_\alpha = Z_{0.01} = 2.326.$$

4) $$Z = \frac{(\bar{x}_2 - \bar{x}_1) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_1}}}$$

$$= \frac{(67.85 - 68.55) - 0}{\sqrt{1.024 \times 10^{-3} + 3.966 \times 10^{-3}}} = \frac{-0.7}{0.070}$$

$$= -10$$

5) conclusion



since the value of z falls under R.R, Reject N.H

Therefore, Australians on an avg taller than Englishmen.

Q) A company claims that alloying reduces resistance of an electric wire by more than 0.05 Ω. To test this claim samples alloyed wire and samples of standard wire are tested which have given the following results.

Type of wire	n	mean resistance	S.D
standard	32	0.136	0.004
Alloyed	32	0.083	0.005

Can the claim be sustained at 0.05 L.O.S.

Ans

1) standard

$n_1 = 32$

$\bar{x}_1 = 0.136$

$s_1 = 0.004$

μ_1

Alloyed.

$n_2 = 32$

$\bar{x}_2 = 0.083$

$s_2 = 0.005$

μ_2

2) N.H $H_0 : \mu_1 - \mu_2 = 0.5\,\Omega$. $\mu_1 > \mu_2$

A.H $H_1 : \mu_1 - \mu_2 > 0.5\,\Omega$ RTT (claim)

3) L.O.S $\alpha = 0.05$

$Z_\alpha = Z_{0.05} = 1.645$

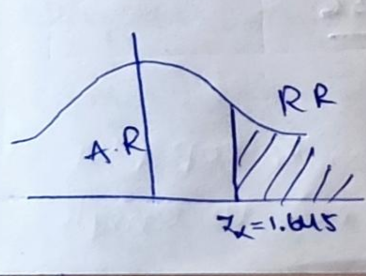
4) Test statistic $\Rightarrow$ $z = \dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$

$= \dfrac{(0.136 - 0.083) - (0.05)}{\sqrt{5 \times 10^{-7} + 7.8125 \times 10^{-7}}} = \dfrac{3 \times 10^{-3}}{1.132 \times 10^{-3}}$

$= \cancel{2.807}$

2.65

5) Conclusion



since the value of z falls under R.R reject NH

$\Rightarrow$ claim is true.

Test of Hypothesis for different 2 means small sample

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$ under NH with $\nu = n_1 + n_2 - 2$ degree of freedom.

where s = pooled standard deviation

$$s = \begin{cases} \sqrt{\dfrac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} & \text{when } s_1 \,\&\, s_2 \text{ are given} \\ \sqrt{\dfrac{\Sigma(x_{1i} - \bar{x})^2 + \Sigma(x_{2i} + \bar{x}_2)^2}{n_1 + n_2 - 2}} & \text{when } s_1 \,\&\, s_2 \text{ not given.} \end{cases}$$

Q: 2 horses A and B were tested acc to the time in seconds to run a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	

Test whether the two horses have the same running capicity

Horse A

$n_1 = 7$

$\bar{x}_1 = \frac{\Sigma x_{1i}}{n_1}$

$= 31.28$

Horse B

$n_2 = 6.$

$\bar{x}_2 = \frac{\Sigma x_{2i}}{n_2}$

$= 28.16.$

$$s = \sqrt{\frac{\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{31.4288 + 26.8336}{11}}$$

$= 2.3014$

2) NH $H_0: \mu_1 = \mu_2$ claim.

A.H $H_1: \mu_1 \neq \mu_2$ (TTT)

3) L.O.S let $\alpha = 0.05$

$\alpha/2 = 0.025$

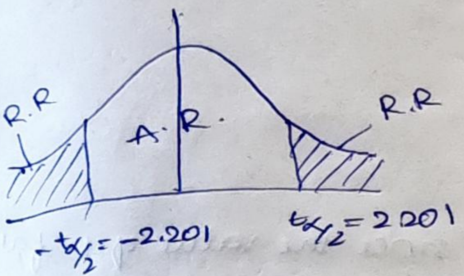
$t_{\alpha/2, \nu} = t_{0.025, 11} = 2.201$

4) $$t = \frac{(31.28 - 28.16) - (0)}{2.3014\sqrt{\frac{1}{7} + \frac{1}{6}}}$$

$$= \frac{3.12}{2.3014(0.5563)}$$

$$= 2.4369$$

Conclusion



since the value of t falls under RR reject N.H.

So two horses don't have the same running capacity.

Q: Samples of 2 types of electric bulbs were tested for length of the life and the following data were obtained

Type-1

$n_1 = 8$

$\bar{x}_1 = 1234$

SD $s_1 = 36$ hrs.

μ_1

Type-2.

$n_2 = 7$

$\bar{x}_2 = 1036$

SD $s_2 = 40$ hrs

μ_2

Is the diff in the means sufficient to warrent that Type-I is superior to Type-2 regarding the length of the life

$$s \text{ (pooled SD)} = \sqrt{\frac{8(36)^2 + 7(40)^2}{8+7-2}}$$

$$s = 40.73$$

NH $H_0: \mu_1 = \mu_2$

A.H $H_1: \mu_1 > \mu_2$ (R.T.T) claim.

3) L.O.S let $\alpha = 0.01$

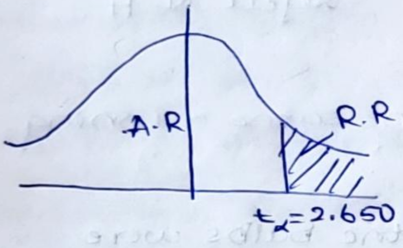
$t_{\alpha,\nu} = t_{0.01, 13} = 2.650$.

4) $t = \dfrac{(1234 - 1036) - 0}{40.73\sqrt{\frac{1}{8} + \frac{1}{7}}}$

$= \dfrac{1234 - 1036}{40.73 \cdot 0.517}$

$= 9.4$

5) Conclusion



since the value of t falls under ~~N.H~~ R.R. Reject N.H

Therefore, Type-1 is superior to Type-2.

Test of Hypothesis for single proportion large population

$$Z = \frac{\bar{p} - P}{\sqrt{\frac{PQ}{n}}} \text{ under N.H}$$

N.H $H_0: P = P_0$

A.H H_1: $P \neq P_0$ T.T.T

$P < P_0$ L.T.T

$P > P_0$ R.T.T

when both sample proportion & pop proportion are given give preference to pop else to sample

Q: In a big city 325 men out of 600 men were found to be smookers. Does this info support conclusion that maj of men in this city are smookers.

1) Let X represents smookers

$x = 325$

$n = 600$ (large)

$\bar{p} = \frac{x}{n} = \frac{325}{600} = 0.541$ (sample proportion).

2) N.H $H_0 : p = 0.5$

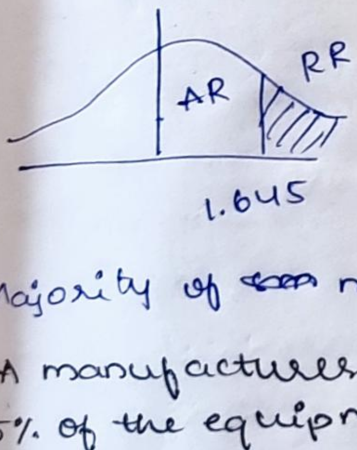
A.H $H_1 : p > 0.5$ (R.T.T) claim.

3) L.O.S let $\alpha = 0.05$

$Z_\alpha = Z_{0.05} = 1.645$

4) $z = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 1.96$

5) Conclusion



since the value of z falls under RR,

Reject N.H.

∴ Majority of ~~men~~ men are smokers.

Q: A manufacturer claimed that atleast 95% of the equipment which he supplied to a factory confirmed to specification. An examination of a sample of 200 pieces of the equipment revealed that 18 were faulty. Test is claimed at 5% level of significance.

1) X = faulty

$x = 18$

$n = 200$ (large)

$\bar{p} = \frac{x}{n} = \frac{18}{200} = 0.09$

P · – pop proportion

2) N.H. H_0: $P = 5\% = 0.05$

A.H H_0:

Test of Hypothesis for 2 proportions large sample.

for small samples, same formula first replace z with t & Z_α with t_α.

Case-1

N.H $H_0: P_1 = P_2$

A.H $H_1: P_1 \neq P_2$

$P_1 < P_2$

$P_1 > P_2$

Q: Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of proposal. Test the hypothesis that the prop of men and women in favour of the proporsal are same. Test at 5% level.

1) men

$n_1 = 400$

$x_1 = 200$

$\bar{P}_1 = \frac{x_1}{n_1} = \frac{200}{400} = 0.5$

P_1

women

$n_2 = 600$

$x_2 = 325$

$\bar{P}_2 = \frac{x_2}{n_2} = \frac{325}{600} = 0.541.$

P_2

2) NH $H_0: P_1 = P_2$ claim

AH $H_1: P_1 \neq P_2$. (TTT).

3) LOS $\alpha = 0.05$

$\alpha/2 = 0.025$

$z_{\alpha/2} = z_{0.025} = 1.96$

4) Test statistic $z = \frac{(\hat{P}_1 - \hat{P}_2) - (P_1 - P_2)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ under NH

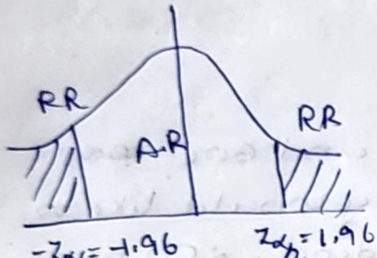
$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 325}{400 + 600} = \frac{525}{1000} = 0.525$

$\hat{q} = 1 - \hat{p} = 0.475$

$$z = \frac{(0.5 - 0.541) - 0}{\sqrt{0.525 \times 0.475\left(\frac{1}{400} + \frac{1}{600}\right)}}$$

$= -1.27$

5) Conclusion



Accept N.H

no diff in proportions.

Q: In a city a 20% of a random sample of 900 school boys have a certain slight physical deffect. In another city B 18.5% of a random sample of 1600 school boys has the same defect. Is the diff b/w proportions significant at 0.05 L.O.S.

1) City A | City B.

$n_1 = 900$ | $n_2 = 1600$

$\bar{P}_1 = 20\% = 0.2$ | $\bar{P}_2 = 18.5\% = 0.185$

P_1 | P_2

$$\hat{P} = \frac{n_1\bar{P}_1 + n_2\bar{P}_2}{n_1 + n_2} = \frac{900 \times 0.2 + 0.185 \times 1600}{900 + 1600.}$$

$= 0.19$

$\hat{q} = 0.81$

2) N.H $H_0 : P_1 = P_2$

A.H $H_1 : P_1 \neq P_2$ claim

Case - II

N.H. $H_0 : P_1 - P_2 = \delta$.

A.H. $H_1 : P_1 - P_2 \neq \delta$

$< \delta$

$> \delta$

Test statistic $z = \dfrac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\dfrac{\bar{P}_1 \bar{q}_1}{n_1} + \dfrac{\bar{P}_2 \bar{q}_2}{n_2}}}$

Q A cigrette manufactruing firm claims that its brand A line of ciggretes outsells its brand B by 8%. If it is found that 42 out of sample of 200 smokers prefer brand A and 18% of another sample of 100 smokers prefer Brand B. Test whether 8% difference is a valid claim.

→

A	B
$n_1 = 200$	$n_2 = 100$
$x_1 = 42$	$x_2 = 18$

$\bar{P}_1 = \dfrac{x_1}{n_1} = \dfrac{42}{200} = 0.21$

2) N.H $H_0 : P_1 - P_2 = 8\% = 0.08$ claim

A.H $H_1 : P_1 - P_2 \neq 0.08$. (TTT)

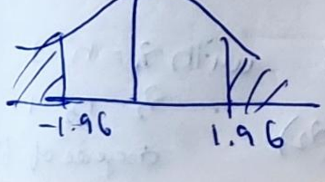
4) Test statistic $z = \dfrac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\sqrt{\dfrac{\bar{P}_1 \bar{q}_1}{n_1} + \dfrac{\bar{P}_2 \bar{q}_2}{n_2}}}$

$= -1.04$

3) LOS $\alpha = 0.05$

$z_{\alpha/2} = z_{0.025} = 1.96$

5)



Accept NH

so his claim is true.

Case-III

Q: In a large populations, there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from a populations

$n_1 = 1200$ $\qquad n_2 = 900$

$Z_{\alpha/2}$ $\qquad x_1 = 30$ $\qquad x_2 = 25$

$P_1 = \frac{x_1}{n_1} = \frac{30}{100} = 0.3$ $\qquad P_2 = \frac{x_2}{n_2} = \frac{25}{100} = 0.25.$

NH. $H_0: \bar{P_1} = \bar{P_2}$

A.H $H_1: \bar{P_1} \neq \bar{P_2}$

$\alpha = 0.05$

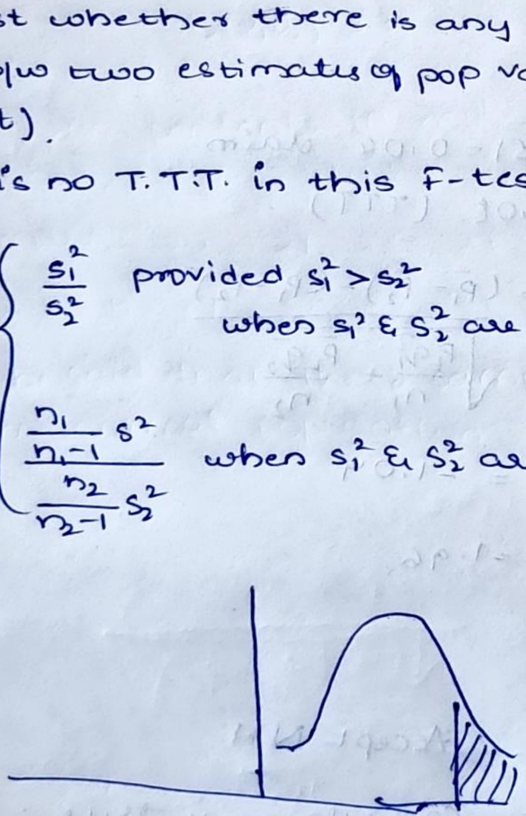
$$z = \frac{P_1 - P_2 - (\bar{P_1} - \bar{P_2})}{\sqrt{\frac{\bar{P_1}\bar{q_1}}{n_1} + \frac{P_2 q_2}{n_2}}}$$

F-Test

(To test whether there is any significant diff b/w two estimates of pop variance we use f-test).

There is no T.T.T. in this F-test.

$$F = \begin{cases} \frac{s_1^2}{s_2^2} \text{ provided } s_1^2 > s_2^2 & \text{when } s_1^2 \text{ \& } S_2^2 \text{ are not given} \\ \frac{\frac{n_1}{n_1-1} s^2}{\frac{n_2}{n_2-1} s_2^2} & \text{when } s_1^2 \text{ \& } S_2^2 \text{ are given.} \end{cases}$$



$F_\alpha(\nu_1, \nu_2)$

with $\nu_1 = n_1 - 1$

$\nu_2 = n_2 - 1$

degree of freedom

Q: In one sample of 8 observations the sum of the squares of deviations of the sample

$n_1 = 8$ (small) $\quad n_2 = 10$

$\Sigma(x_1 - \bar{x}_1)^2 = 84.4 \quad \Sigma(x_2 - \bar{x}_2)^2 = 102.6.$

$$s_1^2 = \frac{\Sigma(x_1-\bar{x}_1)^2}{n_1-1} = \frac{84.4}{7} = 12.06$$

$$s_2^2 = \frac{\Sigma(x_2-\bar{x}_2)^2}{n_2-1} = \frac{102.6}{9} = 11.4$$

$s_1^2 = 12.057 \quad$ N.H $\sigma_1^2 = \sigma_2^2$

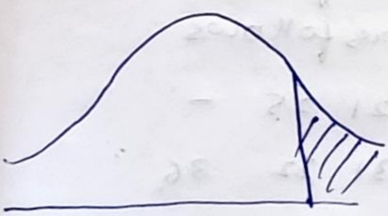
$s_2^2 = 11.4 \quad$ A.H $\sigma_1^2 \neq \sigma_2^2$.

$s_1^2 > s_2^2$

$$F = \frac{s_1^2}{s_2^2} = \frac{12.057}{11.4} = 1.057$$

$F_\alpha(\nu_1, \nu_2) = F_{0.05}(7, 9)$

$= 3.29$



value of F falls under A.R. Acept N.H.

$F_\alpha(\nu_1,\nu_2) = 3.29$

Q Pumpkins were grown under 2 experimental conditions. 2 random samples of 11 and 9 pumpkins show that sample SD of their weights 0.8 and 0.5 respectively. Assuming that their weight distributions are normal, tes the hypothesis that the valience are equal.

$n_1 = 11 \quad n_2 = 9$

$s_1 = 0.8 \quad s_2 = 0.5$

$\sigma_1^2 \quad \sigma_2^2$

$S_1^2 = 0.64$ $\qquad S_2^2 = 0.25$ $\qquad S_1^2 > S_2^2$

2) N.H : $\sigma_1^2 = \sigma_2^2$ claim

A.H : $\sigma_1^2 \neq \sigma_2^2$

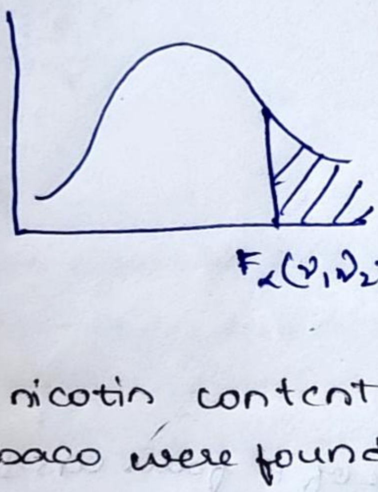
3) LOS, let $\alpha = 5\%$.

$$F_\alpha(\nu_1, \nu_2) = F_{0.05}(10, 8) = 3.35.$$

4) Test statistic

$$F = \frac{\frac{n_1}{n_1 - 1} s_1^2}{\frac{n_2}{n_2 - 1} s_2^2} = \frac{\frac{11}{10}(0.64)}{\frac{9}{8}(0.25)} = 2.503$$

5) conclusion



$F_\alpha(\nu_1, \nu_2) = 3.35$

$F < F_\alpha$, so accept N.H.

Q: The nicotin contents in milligrams in 2 samples of tobaco were found to be as follows

sample A	24	27	26	21	25	–
sample B	27	30	28	31	22	36.

Can it be said that the 2 samples came from the same normal population